Vocational Training Development Institute

School of Applied Technology

Department of ICT

Introduction to Object-Oriented Programming II

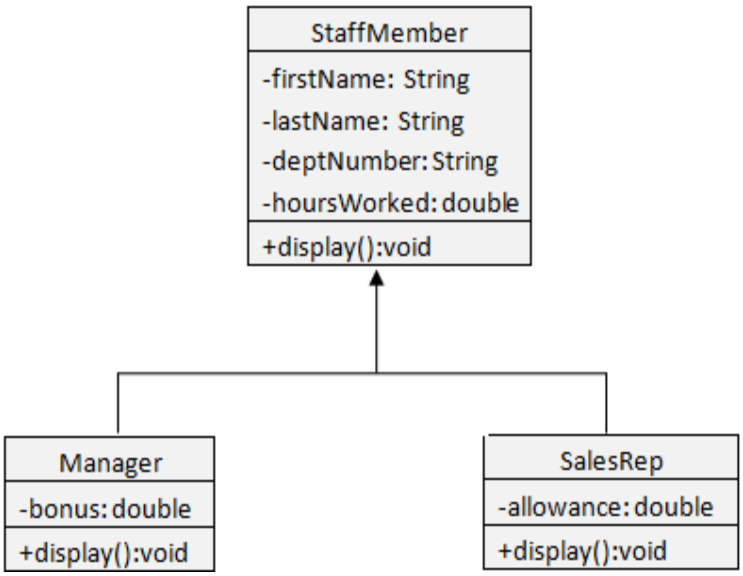
Module No.2

Assignment #2

Due:- **Total marks:** 50 Marks

- *Date*: March 24, 2026 *Lecturer*: Ssyade Gayle
- *Time*: 11:59PM

Create a workspace in suitable location called **OOP2_YourName** to complete the tasks below eg. **OOP2_JohnBrown**.



1) Implement each of the classes in the UML diagram above (***the relationship and access specifiers must be maintained***). Write a ***primary constructor, setters and getters*** for each of the three classes. The ***display()*** method in each class must output/display the value of all the attributes ***(including those inherited***). *Note: write your name and id number at the top of each file as comments.* ***(20 Marks)***

2) Create a file called ***Tester*** for testing the functionality of your application. ***(1 Marks)***

3) Within the ***Tester*** file create an object of the ***base class*** called **obj1** using the ***primary constructor***, then call the object's ***display()*** method. ***(1 Marks)***

4) Create a list of 5 SalesRep using an array list. ***(4 Marks)***

5) Use an iterator to display the contents of the list created above. ***(2 Marks)***

6) Within the ***Tester*** file ***demonstrate polymorphism*** using all the classes in the UML diagram. When your program runs there should be evidence to show that polymorphism has occurred. ***(8 Marks)***

7) Create an ***abstract*** method called ***calculateSalary()*** in the ***base class.*** This method should calculate and display the salary for each staff member, as well as their *firstname*, *lastName* and *DeptNumber*. After doing this it will cause errors to appear in your program when you try to run it. Fix the program by adding or commenting out lines, but do not delete any code. Your program should run correctly now. ***(10 Marks)***
   - *For a **Manager,** salary = ((hoursWorked * 2500) + bonus).*
   - *For a **Sales Rep**, salary = ((hoursWorked * 1500) + allowance).*

8) Use the ***calculateSalary()*** method to display the salary for a ***salesRep*** and a ***Manager*** in the tester file you created above. ***(4 Marks)***